JIMWLK evolution terms from genuine three ρ's

Corrections (blue = new, red = remove; see Evolution_check.pdf):

- Group I, Row II: the second (logarithmic) term has no z left: $\int_{x,y,x',z} \to \int_{x,y,x'}$.
- Group I, Row III, d^3N_2: the first two terms have the wrong sign (in Group_I.tex the bracket has the prefactor $-\frac{1}{(2\pi)^3}$). With the right signs they cancel the fourth and fifth terms exactly (same kernels). The remaining edge-kernel term is $\mathcal{D}$ (fix F3): $U^{bc}(z) \to U^{bc}(w)$ and kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$.
- Group I, Row III, rewritten d^3N_2: as written it is the $[(x-w)^2+(x-z)^2]$ term, which cancels inside Row III ($+\log-\log$). It must be $\mathcal{D}$, which is what your list "All three ρ term with c.c." already uses.
- Group I, Row V: comma in the measure, $\int_{w,z,y,x',y'w'} \to \int_{w,z,y,x',y',w'}$.
- Group II, Row I, rewritten form: four terms read $\mathcal{K}\cdot\mathcal{K}\cdot\mathcal{K}\cdot\mathcal{K}$; it is $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$.
- "From four ρ's": only $\log(k^+/\Lambda)$ belongs in this (JIMWLK) file; the $\log(\vee/k^+)$ part of these terms goes to the BFKL list, where it cancels too. Your cancellation section already uses $\log(k^+/\Lambda)$ only.
- Genuine terms, Group I Row II, d^3N_2: the closing `\Bigg]` is missing.
- Genuine terms, Group III Row III: two terms start with `\-`, which prints nothing; it is a minus sign.
- Everything else is correct (Evolution_check.pdf, Tables 2–4).

1 Group I (with c.c.)

1.1 Row I

Only 4 ρ's

1.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[U^{ab}(w)\mathbb{A}_i^{(3)\,b}(k^+,-w) - \bar{\mathbb{A}}_i^{(3)\,a}(k^+,-w)\right]$$

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{hbc}}{32\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.\\ &\left.-\frac{(y-w)^i}{(y-w)^2}\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]\\ &-\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{abc}}{32\pi^4}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2}\right.\\ &\left.-\frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right]\end{aligned}\tag{1.1}$$

1.3 Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[-\bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+,w;p^+,-z)\bar{\mathbb{A}}_j^{(1)\,b}(p^+,-z) + \bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+,w;p^+,-z)U^{bc}(z)\mathbb{A}_j^{(1)\,c}(p^+,-z)\right]$$

We write it in two parts:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2} \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{1.2}$$

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ & -+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ & -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\,\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2\,(y-w)^2\,(x-z)^2\,(w-z)^2}\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(zw)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ & -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ & +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}\end{aligned} \tag{1.3}$$

We write it as follows:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,w,w',x'} & e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\end{aligned} \tag{1.4}$$

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{abd}}{168\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,w,w',x'} & e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-w)^2-(x-z)^2\right]^2}\\ & \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\\ & \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(zw)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\end{aligned} \tag{1.5}$$

1.4 Row IV

$$\begin{aligned}\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z & \Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\Big]\times\\ & \Big[U^{ac}(w)U^{bd}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\mathbb{B}_{2ji}^{dc}(k^+,-w;p^+,-z) - \bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{B}}_{2ji}^{ba}(k^+,-w;\,p^+,-z)\Big]\end{aligned}$$

$$\begin{aligned}\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} & e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\Bigg[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\\ & +\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\Bigg]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\\ & \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]\end{aligned} \tag{1.6}$$

We can write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\Big[-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z) + \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big] \tag{1.7}$$

1.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_{x,z}\Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\Big]\times \Big[-\bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+,w-z;p^+,z-x)\bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+,-z;p^+,-x) + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+,w-z;p^+,z-x)\,U^{db}(x)\,U^{ce}(z)\,\mathbb{B}^{be}_{2jk}(k^+-p^+,-z;p^+,-x)\Big]$$

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{z,w,y,x',y',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\frac{(x'-w')\cdot(y-w)\,(w-z)\cdot(y'-z)}{(x'-w')^2\,(w-z)^2\,(y-w)^2\,(y'-z)^2} \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(z)\,U^{ce}(w)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]$$
$$-\frac{1}{(2\pi)^3}\int_{w,z,y,x',y',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\frac{(x'-w')\cdot(y'-w)\,(z-w)\cdot(y-z)}{(x'-w')^2\,(z-w)^2\,(y-z)^2\,(y'-w)^2} \times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(w)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big] \tag{1.8}$$

We can write this as (after some simplification)

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w) + \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w) + \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y) - \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\Big] \tag{1.9}$$

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z) - \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x) - \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z) + \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\Big] \tag{1.10}$$

2 Group II

2.1 Row I

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{c'a}(w') - \bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{bc'}(z)\Big]\times \Big[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]$$
$$+\,\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w')\cdot(x'-w)\,(z-w')\cdot(y-z)}{(z-w')^2\,(x-w')^2\,(x'-w)^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w')\cdot(x'-w)}{2\,(x-z)^2\,(x-w')^2\,(x'-w)^2\,(y-z)^2}\right]\times\Big[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\Big]\times$$
$$\Big[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\Big] \tag{2.1}$$

We write it as follows:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^l(y)\,\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^l(y)\,\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{ad}(w')\,U^{de}(w)$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^h(x)\,\rho^l(y)\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{ad}(w')\,U^{de}(w)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^h(x)\,\rho^l(y)\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\Big] \tag{2.2}$$

2.2 Row II

Only 4 ρ's

2.3 Row III

Only 4 ρ's

2.4 Row IV

Only 2 ρ's

3 Group III (with c.c.)

3.1 Row I

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big[\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)U^{c'b}(z)$$
$$-\,\mathbb{A}_i^{\dagger\,(1)\,c'}(k^+,-w')U^{bd'}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,d'}(p^+,-z)U^{c'a}(w')\Big]\times\Big[U^{ac}(w)\mathbb{B}_{2ji}^{bc}(k^+,-w;p^+,-z)-U^{\dagger\,bc}(z)\bar{\mathbb{B}}_{2ji}^{ca}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.1}$$

We can write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\Big[-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$$
$$+\,\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\times\Big[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.2}$$

3.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big[\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')U^{bc'}(z) - \mathbb{A}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\Big]\times$$
$$\Big[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\times\Big[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.3}$$

We write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\Big[-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$$
$$+\,\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\times\Big[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.4}$$

3.3 Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}_{k'}^{\dagger\,(1)\,e'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)$$
$$-\,\bar{\mathbb{A}}_{k'}^{\dagger\,(1)\,c'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)$$
$$-\,U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Big]$$

We write it in two parts:

3.3.1 First part:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big] \tag{3.5}$$

We write it as follows:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,x',w',z,y,x} e^{-ik\cdot(w'-w)}\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)$$
$$\Big[U^{e'c'}(w')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ac}(w)\,\{\rho^c(x),\rho^b(y)\} - U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\big] \tag{3.6}$$

3.3.2 Second part:

Only 2 ρ's

3.4 Row IV

Only 4 ρ's

3.5 Row V

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)$$
$$-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_{j}(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_{i}(k^+,-w)\Big]$$

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-z)\,(x'-w')\cdot(x-w)}{(w'-z)^2\,(x'-w')^2\,(x-w)^2\,(y-z)^2}$$
$$\Big[U^{e'c'}(w')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(y)\,U^{ad}(x)\,\rho^e(y)\rho^d(x) - U^{ad}(w)\,U^{ce}(y)\,\rho^e(y)\rho^d(x)\big] \tag{3.7}$$

We can write it as follows:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$$
$$\Big[U^{e'c'}(w')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(y)\,U^{ad}(x)\,\rho^e(y)\rho^d(x) - U^{ad}(w)\,U^{ce}(y)\,\rho^e(y)\rho^d(x)\big] \tag{3.8}$$

3.6 Row VI

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)$$
$$-\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\mathbb{A}^{(1)\,b}_{j}(p^+,-z)\Big]$$

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-z)\,(x'-w')\cdot(x-w)}{(w'-z)^2\,(x'-w')^2\,(x-w)^2\,(y-z)^2}$$
$$\Big[U^{e'c'}(w')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,\rho^d(x)\,\rho^e(y) - U^{ad}(x)\,\rho^d(x)\,\rho^b(y)\big] \tag{3.9}$$

We can write it as follows:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$$
$$\Big[U^{e'c'}(w')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,\rho^d(x)\,\rho^e(y) - U^{ad}(x)\,\rho^d(x)\,\rho^b(y)\big] \tag{3.10}$$

4 All three ρ term with c.c.

4.1 From four ρ's

$$\frac{d^3N_{a\,\mathrm{NLO}}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x')\,\alpha$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x)\,\alpha'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w)\,\alpha'\Big] \tag{4.1}$$

$$\frac{d^3N_{b\,\mathrm{NLO}}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)\,\alpha\Big] \tag{4.2}$$

4.2 Genuine terms

4.2.1 Group I, Row II

We write it in two parts:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\alpha$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\beta$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)\,\beta$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,\beta$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\beta\Big] \tag{4.3}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\alpha'$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,\beta'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,\beta'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\beta'$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\beta'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,\beta'$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,\beta'$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\beta'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\beta'\Bigg] \tag{4.4}$$

4.2.2 Group I, Row III

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\alpha'$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)\,\beta$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\beta$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)\,\beta$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\beta$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')\,\beta'$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\beta'$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')\,\beta'$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\,\beta'\Bigg] \tag{4.5}$$

4.2.3 Group I, Row IV

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\alpha \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\alpha' \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y)\,\alpha' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x)\,\beta \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\beta' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\beta' \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')\,\beta' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\,\beta'\Bigg]
\end{aligned}
\tag{4.6}
$$

4.2.4 Group I, Row V

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\beta \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\,\beta' \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\beta' \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')\,\beta' \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')\,\beta'\Bigg]
\end{aligned}
\tag{4.7}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\,\beta \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\beta \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\beta \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\,\beta' \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\beta' \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')\,\beta' \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\,\beta'\Bigg]
\end{aligned}
\tag{4.8}
$$

4.2.5 Group II, Row I

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,\alpha \\
&+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\,\alpha \\
&+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)\,\alpha \\
&-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,\beta \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\,\beta \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)\,\beta \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)\,\alpha' \\
&-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\,\alpha' \\
&-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\,\alpha' \\
&+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\,\alpha' \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)\,\beta' \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\,\beta' \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)\,\beta' \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\,\beta'\Bigg]
\end{aligned}
\tag{4.9}
$$

4.2.6 Group III, Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\,\alpha \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\,\alpha \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\,\beta \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\,\beta \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\alpha' \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\alpha' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\beta' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\beta'\Bigg]
\end{aligned}
\tag{4.10}
$$

4.2.7 Group III, Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,\alpha \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\,\alpha \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,\beta \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\,\beta \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w')\,\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\,\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')\,\alpha' \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\,\alpha' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')\,\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\,\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')\,\beta' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\,\beta'\Bigg]
\end{aligned}
\tag{4.11}
$$

4.2.8 Group III, Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^h(x'),\rho^e(y)\right\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\beta$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^e(x'),\rho^h(y)\right\}\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\beta$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^h(x'),\rho^d(y)\right\}\rho^e(x)\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(w')\,\beta$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^e(x'),\rho^h(y)\right\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\beta$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\left\{\rho^h(x),\rho^e(y)\right\}U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\beta'$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\left\{\rho^e(x),\rho^h(y)\right\}U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\beta'$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\left\{\rho^h(x),\rho^d(y)\right\}U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w)\,\beta'$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\left\{\rho^e(x),\rho^h(y)\right\}U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\beta'\Bigg] \tag{4.12}$$

4.2.9 Group III, Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\,\beta$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x)\,\beta'$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)\,\beta'\Bigg] \tag{4.13}$$

4.2.10 Group III, Row VI

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^h(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x')\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x')\,\beta$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(y)\,\rho^h(x')\,\rho^e(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{db}(z)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x)\,\beta'$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\,\beta'\Bigg] \tag{4.14}$$

5 Cancellation of all three ρ's

5.1 α terms

5.1.1 Two Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$$
$$\times\Bigg[-\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w)-\frac{1}{2}\left\{\rho^e(x'),\rho^b(y)\right\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$$
$$+\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x')+\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\Bigg]=0 \tag{5.1}$$

5.1.2 Three Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$$
$$\times\Bigg[-\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)+\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)$$
$$-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)+\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)$$
$$+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x)-\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)$$
$$+\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)-\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)$$
$$+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)-\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$$
$$-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)-\frac{1}{2}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$$
$$+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)+\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)$$
$$+\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)-\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)$$
$$-\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)+\frac{1}{2}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\Bigg]=0 \tag{5.2}$$

5.1.3 Four Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$$
$$\times\Bigg[+\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)+\frac{1}{2}\left\{\rho^h(x'),\rho^l(y)\right\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)$$
$$-\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)-\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w)$$
$$-\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)+\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\Bigg]=0 \tag{5.3}$$

5.2 α' terms

5.2.1 Two Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$$
$$\times\Bigg[-\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x)-\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')$$
$$+\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w)+\frac{1}{2}\,\rho^c(x')\left\{\rho^e(x),\rho^b(y)\right\}U^{ad}(w')\,U^{de}(w)\Bigg]=0 \tag{5.4}$$

5.2.2 Three Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\ &\times\Big[\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\\ &+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\\ &-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y)+\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\\ &-\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\\ &-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\\ &+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y)+\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\\ &-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')-\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\\ &-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)+\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\\ &+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)-\frac{1}{2}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\Big]=0\end{aligned}\tag{5.5}$$

5.2.3 Four Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\ &\times\Big[-\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)-\frac{1}{2}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\\ &+\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')+\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w')\\ &+\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)-\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\Big]=0\end{aligned}\tag{5.6}$$

5.3 β terms

5.3.1 Two Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\ &\times\Big[-\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)+\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)-2\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\\ &+\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)-\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)+\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\Big]=0\end{aligned}\tag{5.7}$$

5.3.2 Three Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[+\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x)\\
&+\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)-\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\\
&+\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&-2\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x)+\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')-2\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x')\\
&-\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x)\\
&-\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)+\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\\
&+2\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)+2\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)-\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')+2\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)-2\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)\\
&-\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)+\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\\
&-2\,\rho^e(y)\,\rho^h(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x')+\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\\
&-\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)+\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\\
&-\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)-\big\{\rho^h(x'),\rho^d(y)\big\}\rho^e(x)\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(w')\\
&+2\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)+2\,\rho^d(y)\,\rho^h(x')\,\rho^e(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{db}(z)\Big]=0\end{aligned}\tag{5.8}$$

5.3.3 Four Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[-2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)-\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)\\
&+2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)+2\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)\\
&+2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)-2\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\Big]=0\end{aligned}\tag{5.9}$$

5.4 β' terms

5.4.1 Two Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[+\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')-\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\\
&+2\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')-\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')+\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\\
&-\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)\Big]=0\end{aligned}\tag{5.10}$$

5.4.2 Four Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[+2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)+\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\\
&-2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')-2\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')\\
&-2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)+2\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\Big]=0\end{aligned}\tag{5.11}$$

5.4.3 Three Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\\
&-\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')+\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\\
&-\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&+2\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')-\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x)+2\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x)\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')\\
&+\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')-\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\\
&-2\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')-2\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')+\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)-2\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x)\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')+2\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)\\
&+\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')-\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\\
&+2\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)-\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\\
&+\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')-\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\\
&+\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)+\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w)\\
&-2\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')-2\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\Big]=0\end{aligned}\tag{5.12}$$